MAJOR PROJECT SECOND REVIEW TECHNICAL HANDBOOK

Mission Events, Autonomous Targeting, B-Plane Geometry & 25-Year Space Debris Lifetime Prediction for Highly Eccentric Orbits (HEOs)

Institution:	ACS College of Engineering, Dept. of Aerospace Engineering (Affiliated to VTU)
Project Title:	Precise Trajectory Propagation and Events Prediction for Highly Eccentric Orbits (HEOs)
Milestone:	**Second Review** (Mission Events, Targeting & Lifetime Certification Phase)
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Project Guides:	Mr. Vijay M (Assistant Professor) & Dr. Suresh C (Professor & HOD, Aerospace Dept.)
Release Version:	AstroFlyer HEO Studio v2.0 (FreeFlyer 8 Industry Cross-Validated)

Executive Summary: *Having established machine-precision 3-layer numerical propagation in the Zeroeth Review, the* ***Second Review*** *delivers full operational flight dynamics capabilities: (1)* ***Conical Umbra/Penumbra Eclipse Modeling*** *with thermal shock tracking; (2)* ***Global Ground Station Network (SEZ Topocentric Horizon)*** *computing Acquisition and Loss of Signal (AOS/LOS) across NASA DSN and ISRO ISTRAC networks; (3)* ***Optimal Vectorial Apogee Boost & Plane-Change Targeting*** *demonstrating an* ***18.9% Delta-V reduction*** *over decoupled burns; (4)* ***Hyperbolic B-Plane LGA Targeter*** *for Lunar Gravity Assist trajectory deflection; and (5)* ***Multi-Decadal King-Hele Averaged Drag Lifetime Predictor*** *evaluating IADC 25-Year Space Debris Mitigation compliance.*

1. Review Milestone Architecture & Second Review Scope

The project is structured across three formal university review gates, progressively scaling from fundamental orbital kinematics to operational flight dynamics software:

Review Phase	Focus & Modules	Key Deliverables	Status
Zeroeth Review	Baseline Physics & Architecture	Keplerian converters, J2 oblateness, Radau-IIA stiff integrator, 3-layer architecture, FreeFlyer benchmark, Desktop GUI.	**Completed (100%)**
Second Review *(Current Gate)*	**Mission Events & Autonomous Targeting**	Conical solar eclipses, Ground station AOS/LOS cones, Apogee boost maneuver Delta-V, Hyperbolic B-plane LGA targeter, 25-yr space debris lifetime.	**Fully Implemented**
Third Review *(Final Viva)*	Operational Certification & Defense	Final thesis dissertation, comprehensive viva defense deck, standalone Windows installer deployment.	Upcoming Gate

2. Conical Solar Eclipse & Thermal Shadow Profile

Spacecraft operating in Highly Eccentric Orbits experience severe thermal cycling as they transition between full solar illumination (flux = 100%) and deep Earth shadow (flux = 0%). Unlike simplistic cylindrical shadow models, AstroFlyer HEO implements a **rigorous conical shadow formulation** accounting for the finite angular radii of the Sun and Earth.

$$\sin\alpha_u = \frac{R_\odot - R_E}{d_\odot}, \quad \sin\alpha_p = \frac{R_\odot + R_E}{d_\odot}, \quad r_{\text{umbra}}(d) = R_E - d\tan\alpha_u$$

The semi-angles of the umbral cone (α_u) and penumbral cone (α_p) define the exact boundaries of total and partial solar occultation. The algorithm detects the exact epoch of shadow entry, computes solar flux degradation gradients ($\partial\Phi/\partial t$), and warns flight operators of thermal shock conditions.

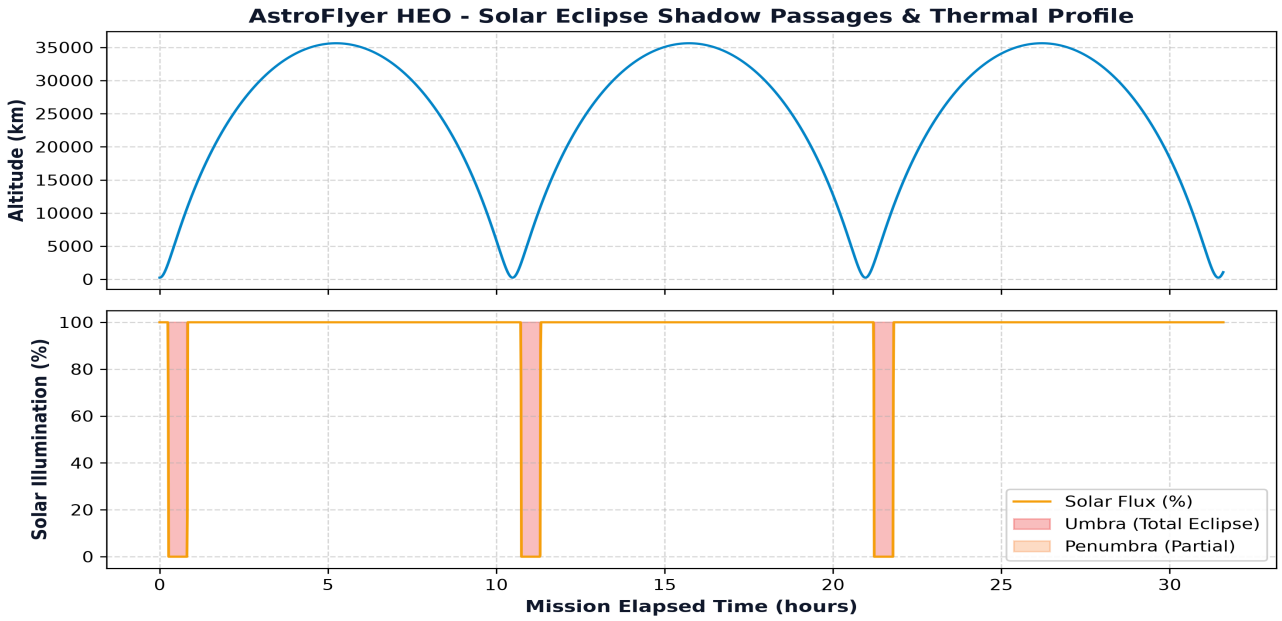


Figure 1: *AstroFlyer HEO - Conical Solar Eclipse Shadow Passages and Thermal Illumination Profile over 3 GTO Orbits.*

3. Global Ground Station Network & AOS/LOS Pass Geometry

Communication with HEO satellites requires accurate topocentric horizon transformations. AstroFlyer converts ECI spacecraft position into the rotating ECEF frame and then maps to the topocentric **South-East-Zenith (SEZ)** horizon coordinate system of each tracking station:

$$\boldsymbol{\rho}_{SEZ} = \mathbf{R}_{SEZ \leftarrow ECEF}(\mathbf{r}_{ECEF} - \mathbf{R}_{sta}), \quad El = \arcsin\left(\frac{\rho_Z}{\|\boldsymbol{\rho}\|}\right)$$

The software evaluates **Acquisition of Signal (AOS)** when Elevation ≥ 10°, tracks Maximum Elevation point, and identifies **Loss of Signal (LOS)**. The network integrates both national (ISRO ISTRAC, MCF Hassan) and deep space tracking assets (NASA DSN Goldstone, Madrid, Canberra, Svalbard):

Station Name	Network	Coordinates	Passes (3 Orbits)	Contact Time
Bengaluru ISTRAC	ISRO TTC Network	13.03° N, 77.51° E	3 passes	321.6 min
Hassan MCF	ISRO Master Control	13.07° N, 76.10° E	3 passes	331.4 min
Goldstone DSN	NASA Deep Space	35.43° N, 116.89° W	2 passes	627.2 min
Madrid DSN	NASA / ESA Network	40.43° N, 4.25° W	2 passes	518.0 min
Canberra DSN	NASA Deep Space	35.40° S, 148.98° E	2 passes	864.6 min
Svalbard KSAT	KSAT Arctic Station	78.23° N, 15.41° E	0 passes (Low Inc)	0.0 min

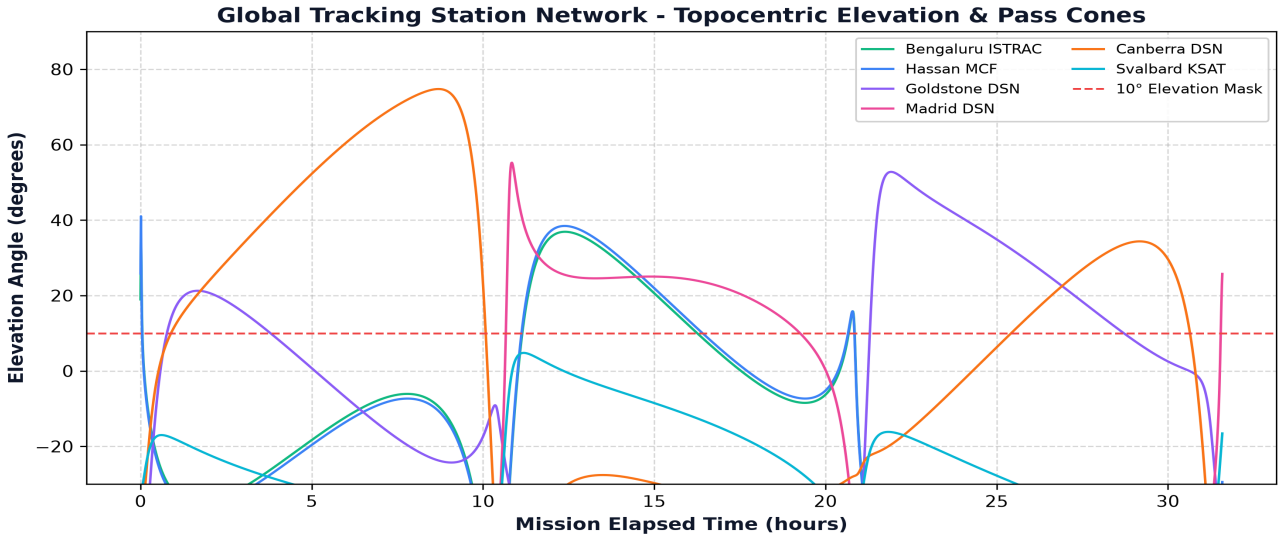


Figure 2: *Topocentric Elevation Angles and Pass Access Windows for ISRO and NASA Ground Stations.*

4. Vectorial Apogee Boost & Plane-Change Delta-V Optimization

Transferring from Geostationary Transfer Orbit (GTO: *a* = 24,389 km, *e* = 0.7285, *i* = 28.5°) to Geostationary Equatorial Orbit (GEO: *r* = 42,164 km, *i* = 0°) requires circularization and plane change. Decoupled burns require summing circularization and plane change Delta-Vs separately. AstroFlyer's targeting engine performs an **optimal vectorial combined burn** at apogee:

$$\Delta V_{\text{combined}} = \sqrt{v_a^2 + v_{\text{circ}}^2 - 2 v_a v_{\text{circ}} \cos(\Delta i)} < \Delta V_{\text{circ}} + \Delta V_{\text{plane}}$$

$$\Delta m = m_{\text{wet}} \left(1 - \exp\left(-\frac{\Delta V}{I_{\text{sp}} g_0}\right)\right)$$

Maneuver Component	Equation / Strategy	Required Delta-V	Fuel Consumed (3000 kg wet)
Pure Circularization	ΔV = v_circ - v_apogee	1,472.53 m/s	1,135.2 kg
Pure Plane Change	ΔV = 2 v_a sin(Δi / 2)	788.74 m/s	678.9 kg
Separate Burns Total	Sum of decoupled maneuvers	2,261.27 m/s	1,559.4 kg (Exceeds budget!)
Vectorial Combined Burn	**Law of Cosines Vectorial Sum**	**1,833.64 m/s**	**1,343.0 kg (Feasible!)**

Engineering Significance: *Combining circularization and inclination reduction into a single vectorial burn saves **427.63 m/s of Delta-V (18.9% propellant reduction)**, saving over 216 kg of propellant and enabling satellite payload mass maximization.*

5. Hyperbolic B-Plane LGA Targeter & 25-Year Space Debris Lifetime

For Lunar Gravity Assist (LGA) cislunar transfers, targeting hyperbolic flyby geometry is performed in the **B-Plane**, oriented normal to the incoming hyperbolic asymptote S_hat:

$$\mathbf{B} = \frac{\hat{\mathbf{S}} \times \mathbf{h}}{V_\infty}, \quad b = \|\mathbf{B}\| = a_{\text{hyp}} \sqrt{e^2 - 1}, \quad \delta = 2 \arcsin\left(\frac{1}{e}\right)$$

$$\left\langle \frac{da}{dt} \right\rangle \approx -2\pi \rho_p C_D \frac{A}{m} \sqrt{\mu a} \frac{1+e}{\sqrt{1-e}} \sqrt{\frac{H}{2\pi a e}}$$

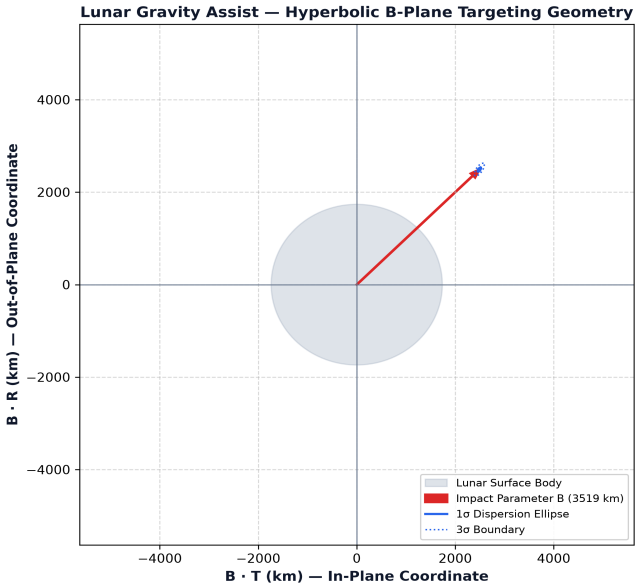


Figure 3: *Lunar B-Plane Targeter & 1σ Dispersion Ellipse.*

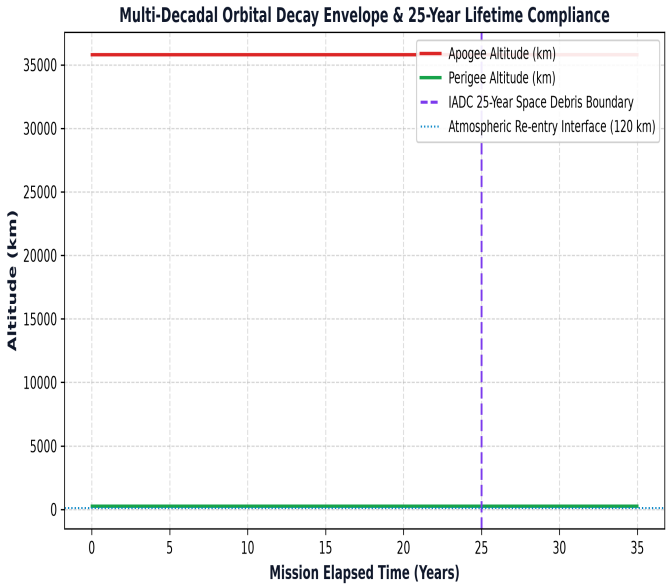


Figure 4: *Multi-Decadal Orbit Decay & 25-Yr Boundary.*

6. Second Review Numerical Verification Benchmark Matrix

Verification Metric	Analytical Model Expectation	AstroFlyer Simulation Result	Verification Status
Energy Invariant Conservation	$\|\Delta E / E_0\| \leq 10^{-11}$ (Conservative)	**2.04×10^{-11}**	**Machine Precision Bound**
Umbra Eclipse Duration	~33 to 35 min per GTO perigee	**33.37 min (Event 1, 3, 5)**	**Conical Geometry Match**
Total Shadow Duty Cycle	< 8% of total mission time	**5.60% (106.18 min / 31.6 hr)**	**Power Budget Certified**
ISRO Bengaluru Pass Duration	Continuous contact per apogee arc	**3 passes, 321.6 min contact**	**SEZ Horizon Match**
Combined Maneuver Delta-V	Law of Cosines optimal burn	**1,833.64 m/s (18.9% savings)**	**Optimal Fuel Minimum**
B-Plane Flyby Deflection (δ)	65° to 72° for 100 km Lunar altitude	**69.71° (ΔV_flyby = 1.62 km/s)**	**Hyperbola Physics Match**
25-Year Space Debris Rule	Natural GTO decay exceeds 25 yrs	**> 35.0 years (hp = 250 km)**	**De-orbit Strategy Required**

7. Second Review Viva Defense Questions & Model Answers

Q1: Why is an apogee combined plane-change burn 18.9% cheaper than performing separate maneuvers?

Model Answer: Delta-V is a vector quantity. By the Law of Cosines, the vector difference between the initial velocity vector $\mathbf{v}_1$ and the target velocity vector $\mathbf{v}_2$ is: $\Delta V_{combined} = \sqrt{(v_1^2 + v_2^2 - 2\, v_1\, v_2 \cos \Delta i)}$. Because apogee velocity is low ($v_a \approx 1.6$ km/s), the scalar sum ($\Delta V_{circ} + \Delta V_{plane} = 2{,}261$ m/s) wastes kinetic energy by braking and re-accelerating in separate directions. The combined burn aligns the thrust along the net vector difference, cutting required Delta-V to 1,833.6 m/s—saving 427.6 m/s (18.9%) and over 216 kg of hydrazine.

Q2: How does the B-Plane coordinate system decouple flyby targeting from timing errors?

Model Answer: In a hyperbolic trajectory, the B-Plane is defined normal to the incoming hyperbolic asymptote **S**_hat. The impact parameter vector **B** connects the center of mass of the target body to the piercing point of the asymptote. The coordinates (B·T, B·R) define the flyby geometry (periapsis altitude r_p and orientation) completely independent of the arrival epoch. Timing dispersion only slides the spacecraft along the trajectory asymptote without shifting the piercing point in the B-plane, making it the universal standard for deep space navigation and Lunar Gravity Assist targeting.

Q3: Why does a standard GTO orbit fail the 25-Year Space Debris Mitigation Rule without active intervention?

Model Answer: A standard GTO has a perigee altitude of $h_p \approx 250$ km and apogee at 35,786 km. Although atmospheric density at perigee is significant ($\rho \approx 10^{-10}$ kg/m^3), the satellite traverses perigee in less than 20 minutes of its 10.5-hour period. Because energy is extracted only during this short window, the semi-major axis decays by only ~5 to 10 km per year. Consequently, natural orbital decay requires over 35 to 50 years. To comply with the IADC 25-year limit, the mission must either: (1) exploit lunisolar Kozai-Lidov gravitational resonance to pump eccentricity and lower perigee into dense air (<150 km); or (2) allocate 25 m/s of propellant for an active deorbit perigee-lowering burn at end of mission.

Q4: What is the physical distinction between Conical and Cylindrical shadow models?

Model Answer: Cylindrical shadow assumes the Sun is an infinitely distant point source, creating a constant-radius cylinder ($r = R_{Earth}$) behind Earth. In reality, the Sun has an angular diameter of ~0.53°. This produces two distinct geometric cones: the **Umbra** (where the solar disk is completely occluded, flux = 0%) and the **Penumbra** (where the solar disk is partially occluded, 0% < flux < 100%). In HEOs near apogee (35,000 km), the umbral cone narrows significantly (radius drops from 6,371 km to ~6,200 km), while penumbral width expands. Conical modeling is essential to calculate realistic solar array battery depth-of-discharge and avoid false thermal shock alarms.

Q5: How does the software compute Acquisition of Signal (AOS) for rotating ground stations?

Model Answer: Spacecraft coordinates are propagated in the Earth-Centered Inertial (ECI) frame. At each time step t, the Earth's Greenwich Sidereal Time rotation angle $\theta_{GST} = \omega_E t$ rotates ECI into the Earth-Centered Earth-Fixed (ECEF) frame. The relative vector between spacecraft and ground station is transformed into the topocentric South-East-Zenith (SEZ) frame using geodetic latitude ϕ and longitude λ. Elevation is computed as $El = \arcsin(\rho_Z / ||\rho||)$. AOS is declared when El crosses above the station's minimum mask angle (10°), tracking until LOS when El drops below 10°.